Power for Linear Regression

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Introduction

Power analysis for linear regression determines the sample size required to detect either an overall non-zero R^2 (via the omnibus F-test of the regression) or the incremental R^2 contributed by adding one or more predictors to a baseline model. Both questions are central to the design of observational and experimental studies that use multiple regression for inference: the omnibus power calculation tells you whether the model as a whole has a chance to be detected, and the incremental power calculation tells you whether a specific block of predictors of scientific interest will be detectable above and beyond control variables. Cohen's f^2 effect size unifies these calculations into a single non-central F-distribution framework.

Prerequisites

A working understanding of multiple linear regression, the relationship between R^2 and the omnibus F-statistic, and Cohen's f^2 as the standardised effect-size measure for regression.

Theory

Cohen's f^2 for the omnibus regression test is

$$f^2 = \frac{R^2}{1 - R^2},$$

and for the incremental contribution of a new block of predictors,

$$f^2 = \frac{R^2_{\text{full}} - R^2_{\text{reduced}}}{1 - R^2_{\text{full}}}.$$

Conventional benchmarks are 0.02 (small), 0.15 (medium), and 0.35 (large). The omnibus or partial F-test follows a non-central F-distribution under H_1 with non-centrality $\lambda = f^2(u + v + 1)$, where u is the numerator df (number of predictors tested) and v is the residual df. Power is the probability that this non-central F exceeds the critical value at α.

Assumptions

The standard linear-regression assumptions (linearity, homoscedasticity, Normal residuals, independent observations), the number of predictors in the full model and the incremental block are pre-specified, and the effect size f^2 is chosen on the basis of the smallest scientifically meaningful incremental R^2.

R Implementation

```r
library(pwr)

pwr.f2.test(u = 5, v = NULL, f2 = 0.15, sig.level = 0.05, power = 0.80)

pwr.f2.test(u = 2, v = NULL, f2 = 0.02, power = 0.80)
```

Output & Results

`pwr.f2.test()` returns the residual degrees of freedom v required to achieve target power; the total sample size is $n = v + u + 1$. For a 5-predictor model at medium effect $f^2 = 0.15$, $v = 85$ and $n = 91$; for a small incremental effect $f^2 = 0.02$ with 2 added predictors after 5 baseline predictors, $v = 489$ and $n = 497$ — illustrating how dramatically small incremental effects increase sample-size requirements.

Interpretation

A reporting sentence: "With 5 predictors in the regression and an anticipated $f^2 = 0.15$ (medium-effect convention; equivalent to $R^2 = 0.13$), a total sample of $n = 91$ is required to achieve 80 % power for the omnibus F-test at $\alpha = 0.05$. To detect an incremental $f^2 = 0.02$ from 2 additional predictors of primary interest after adjusting for the 5 baseline covariates, the protocol requires $n = 497$. The trial therefore enrols 550 participants to allow for 10 % attrition." Always state u, f^2, and the inflation factor.

Practical Tips

- Convert anticipated R^2 expectations to f^2 using the formulas above; the conversion is non-linear and a "small R^2" of 0.05 corresponds to $f^2 \approx 0.053$, while a "medium R^2" of 0.13 corresponds to $f^2 \approx 0.15$.
- Small incremental effects ($f^2 < 0.05$) require very large samples to detect; weigh the cost-benefit honestly and consider whether a smaller incremental effect is genuinely scientifically interesting.
- Interactions, polynomial terms, and dummy-coded categorical variables each add to the predictor count u and inflate the standard errors of individual coefficients; the sample-size calculation must reflect the actual model structure.
- Pre-specify the minimum detectable incremental effect of interest in the protocol; avoid post-hoc rationalisation of the chosen f^2 to match the achieved power, which is a form of HARKing.
- Include a 15–20 % buffer above the calculated n for residual non-Normality, minor violations of homoscedasticity, and missing data; the calculated n is the minimum, not a target.
- For prediction-focused studies (rather than inferential studies), consider using a learning-curve or repeated-cross-validation framework to determine the sample size required for stable out-of-sample R^2, which is a different question than power for the omnibus test.

R Packages Used

`pwr::pwr.f2.test()` for canonical f^2-based linear-regression power calculation; `WebPower::wp.regression()` for an alternative interface mirroring popular online calculators; `pwrss::pwrss.f.reg()` for extended regression-power tools; `Superpower` for ANOVA and factorial-design power simulation; `simr` for simulation-based power on regression with complex covariance structures.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Sample size, power, and Quarto reporting
- Power: closed-form and simulation